

Shrinkage: Ridge, Principal Components and the Lasso

Least squares is the best linear unbiased estimator, and under normal errors the best unbiased estimator of any kind. With nearly dependent columns it can still be very bad, because its variance along poorly determined directions swamps everything else. This chapter gives up unbiasedness on purpose, measures estimators by mean squared error, and studies three ways of buying a large cut in variance with a little bias. Ridge regression shrinks every direction of the coefficient space, the weakly determined ones most; principal component regression discards the weakly determined directions; the lasso shrinks coefficients with an absolute-value penalty and sets some exactly to zero. The chapter ends with Stein's paradox: in three or more dimensions a shrinkage estimator beats least squares for every value of the parameter, for an empirical Bayes reason that also explains the other methods.

What you need. [Chapter 1](#) (the singular value decomposition, Theorem 1.8.1), [Chapter 6](#) (projections and the variance inflation identity Eq. 6.6.2), [Chapter 7](#) (the Gauss–Markov theorem, Theorem 7.2.1, and the conjugate normal model, Theorem 7.6.2 and Proposition 7.6.3), [Chapter 10](#) (least squares through the SVD, Theorem 10.4.1 and Proposition 10.4.3), [Chapter 19](#) (the short regression as a biased estimator, Theorem 19.1.5), [Chapter 20](#) (the leave-one-out formulas, Theorem 20.3.1) and [Chapter 26](#) (what collinearity does to variances, Theorem 26.1.1). The James–Stein section uses the noncentral chi-squared distribution of [Chapter 4](#) (Theorem 4.1.2) together with the normal conditional distribution of [Chapter 3](#) (Theorem 3.5.2), which is where the empirical-Bayes reading of the estimator comes from; [Section 27.1](#) uses the covariance of a linear transformation (Theorem 2.2.1, [Chapter 2](#)).

27.-1.1 Roadmap

- [The bias–variance trade-off](#)
- [Ridge regression](#)
- [Principal component regression](#)
- [The lasso](#)
- [Stein's paradox and James–Stein estimation](#)
- [Summary and notes](#)